Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala
(Deemed to be University)
Department of Electrical and Instrumentation Engineering
Auxiliary Examination

B.E	Course Code: UCS 520
Time: 03:00 hrs	Course: Computer Networks
Max Marks: 100	Date: 15/02/2025

Q1.	Compute the fraction of the bandwidth that is wasted on overhead (headers and retransmissions) for protocol on a heavily loaded 50-kbps satellite channel with data frames consisting of 40 bits header and 3960 data bits. Assume that the signal propagation time from the earth to the satellite is 270 msec. ACK frames never occur. NAK frames are 40 bits. The error rate for data frames is 1%, and the error rate for NAK frames is negligible.	**(20)**
Q2.	**a.** What is the role of the Header Checksum field in Internet Protocol version 4 (IPv4) header format? Explain clearly (with appropriate sketches). Consider the 3-way handshake mechanism followed during TCP connection establishment between hosts P and Q. Let X and Y be two random 32-bit starting sequence numbers chosen by P and Q respectively. Suppose P sends TCP connection request message to Q with TCP segment having SYN bit = 1, SEQ number = X, and ACK bit = 0. Suppose Q accepts the connection request. Now write condition to represent the information present in TCP segment header that is sent by Q to P?	**(10)**
	b. Explain the Reverse Address Resolution Protocol (RARP) process in the Ethernet network. Suppose A host with IP address 130.23.43.20 and physical address B2:34:55:10:22:10 has packet to send to another host with IP address 130.23.43.25 and physical address A4:6E:F4:59:83:AB. The two hosts are on the same Ethernet network. Show ARP request and reply packets encapsulated in Ethernet frames.	**(10)**
Q3.	**a).** The following character encoding is used in a data link protocol: A: 01000111; B: 11100011; FLAG: 01111110; ESC: 11100000. Show the bit sequence transmitted (in binary) for the four-character frame: A B ESC FLAG when each of the following framing methods is used: **i.** Character Count, **ii.** Flag bytes with byte stuffing, **iii.** Starting and ending flag bytes with bit stuffing.	**(10)**
	b). Host A sends a UDP datagram containing 8880 bytes of user data to host B over an Ethernet LAN. Ethernet frames may carry data up to 1500 bytes (i.e. MTU = 1500 bytes). The size of the UDP header is 8 bytes, and the size of the IP header is 20 bytes. There is no option field in the IP header. Estimate how many total number of IP fragments will be transmitted and what will be the contents of the offset field in the last fragment. Explain your answer.	**(10)**
Q4.	Two nodes (A and B) are on the same network, and distanced by 500-meter. The two nodes are connected via 3 switches, each with a delay of 5 μs (microsecond). Suppose that both A and B transmit 1500 bytes at the same exact time, and their frames collide. A chooses to retransmit immediately after the jam signal is received, and B chooses to retransmit after 50 μs of receiving the jam signal. Assume the transmission speed is 10 Mbps, the channel is CSMA/CD with backoff intervals that are multiples of, the jam signal takes 10 μs to transmit, and the inter-frame gap is 9.6 μs. Find the time required to deliver A's packet to B.	**(20)**
Q5.	**a).** Consider a TCP connection between a client and a server with the following specifications: RTT (Round Trip Time) is 6 ms, the size of the receiver advertised window is 50 KB, the slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time t = 0. Assume that there are no timeouts and errors during transmission. Then estimate the size of the congestion window (in KB) at time t + 60 ms after all acknowledgments are processed.	**(10)**
	b). Explain the advantages and disadvantages of the Mesh and Ring network topology. Also, elaborate on the five responsibilities of the Presentation Layer of the OSI model.	**(10)**